

Applied Machine Learning Exam Report

Project 1: Customer Churn Prediction in a Subscription-Based Service

DSK 807 Project

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1 Introduction

Customer churn prediction is a key challenge for subscription-based companies, as losing an existing customer is typically more costly than contacting a customer who would otherwise stay. The objective of this project is therefore to identify as many customers at risk of churn as possible, even if this results in contacting some non-churning customers. Using the Telco Customer Churn dataset, machine learning models are trained to classify customers as likely to churn or likely to stay based on customer data.

Because churn customers represent a minority of the population and false negatives are particularly costly, overall accuracy, on its own, can be a misleading metric. Instead, the project prioritizes recall-oriented performance, with the F_2 score used as the main evaluation criterion, while accuracy and ROC–AUC are reported as complementary metrics. This reflects an aggressive churn detection strategy focused on early identification rather than conservative classification.

2 Exploratory Data Analysis

The dataset contains 7,043 observations with 21 features. After removing an irrelevant customer identifier feature, 19 predictors remain alongside the binary target *Churn*. Approximately 27% of customers have churned, resulting in a moderate class imbalance that motivates stratified splits and recall-focused evaluation.

Only three variables are numerical: *tenure*, *MonthlyCharges*, and *TotalCharges*. Tenure is strongly right-skewed, with churn customers concentrated at the lower end, indicating that often churn occurs early in the customer lifecycle. MonthlyCharges shows higher churn rates at higher price levels, while TotalCharges is strongly correlated with tenure and reflects cumulative billing behavior.

The remaining features are predominantly low-cardinality categorical variables describing contracts and service subscriptions. Month-to-month contracts and the absence of add-on services such as technical support or online security are associated with higher churn, whereas most demographic variables show weaker correlation with the exception of the *SeniorCitizen*, and *Dependents* features. A small number of missing values in TotalCharges occur primarily for short-tenure customers and likely reflects limited billing history rather than random missingness.

Overall, the EDA indicates that churn is driven mainly by short tenure, weak contractual commitment, and pricing-related factors. These patterns support a recall-oriented modeling strategy, as failing to identify early-stage churners would undermine the primary business objective.

3 Shallow Learning Approaches

We train and test three different shallow learning models for this classification task. They are Logistic Regression, Random Forest, and XGBoost. We also experiement with ensemmble combinations of all 3 models.

During hyperparameter tuning, three scoring metrics where used for each of three models, resulting in nine total tuned shallow models. The metrics were accuracy, ROC–AUC, and F_β-score with $\beta = 2$. Each metric emphasizes different aspects of model performance, and optimizing them leads to different behaviors, even for the same model. Understanding each of these metrics, and the performance they underline is essential in calibrating the optimal model for the case at hand.

Accuracy is highly intuitive and therefore a universal metric for classification. It is however, poorly suited for imbalanced classification problems as it is biased toward the majority class. In the case of customer churn, we are interested in predicting churn so that it can be minimized and are therefore near exclusively interested in the minority class. Optimizing for accuracy during hyperparameter tuning is therefore not well aligned with our objective as it often results in low recall for the minority class.

ROC–AUC ranks quality across all decision thresholds and is therefore a strong indicator of a models true ability to differentiate instances of the target class. It is also more robust to class imbalances than accuracy. It does not however, reflect performance at a deployed decision threshold and therefore gives an incomplete view of true performance.

F_β explicitly prioritizes recall over precision and , similarly to the decision threshold, is adjustable in coordination with stakeholders. It is more aligned with the goal of identifying the minority class with the tradeoff of increased false positive rates.

In the following subsections we show the results of each of the fine tunings across each of the three metrics.

3.1 Logistic Regression

Logistic Regression is an informative baseline when performing classification as the model is linear, and learning is stable. Using a linear model as a baseline can help you catch overfitting if you don't observe substantial increases in performance for higher capacity models. We therefore deliberately choose to train this model first.

Scoring	Test Accuracy	Test ROC-AUC	F_2
Accuracy	0.80	0.83	0.54
ROC-AUC	0.80	0.84	0.58
F_2	0.69	0.82	0.73

Table 1: Comparison of logistic regression models tuned with different scoring metrics. F_2 optimization achieves positive class recall of 0.87

3.2 Random Forest

Scoring	Test Accuracy	Test ROC-AUC	F_2
Accuracy	0.79	0.83	0.52
ROC-AUC	0.79	0.83	0.53
F_2	0.73	0.83	0.70

Table 2: Comparison of random forest models tuned with different scoring metrics, F_2 optimization achieves positive class recall of 0.78

3.3 XGBoost

Scoring	Test Accuracy	Test ROC-AUC	F_2
Accuracy	0.80	0.84	0.52
ROC-AUC	0.72	0.84	0.72
F_2	0.67	0.84	0.73

Table 3: Comparison of random forest models tuned with different scoring metrics, F_2 optimization achieves positive class recall of 0.87

3.4 Ensemble Methods

Both stacking and voting ensembles of the three aforementioned classifiers were implemented and tested. None of these achieved substantial performance improvement over the basic shallow learners.

4 Deep Learning Approaches

4.1 Feed Forward Neural Network

Feed forward neural network models using various combinations of optimizers, learning rates, batch sizes, regularizations, and other relevant parameters were used to find a model that optimizes accuracy. We observe that the results of the higher capacity, deep learning, models offer only marginal performance increases when compared to the shallow methods. Because of this, and because of the size of the dataset, higher capacity models increase our risk of overfitting, for marginal reward.

4.2 Autoencoder

The purpose of using an autoencoder in this task is dimensionality reduction of the dataset, and to evaluate whether a latent representation of the data is capable of carrying, or boosting the signal within the original data. We test this by retraining our

shallow learners on latent representations, benchmarking their performances against their counterparts, trained on the uncompressed data.

Our results show that dimensionality reduction using autoencoders do not lead to improved accuracy on shallow learners. In fact, for meaningfully small latent spaces, accuracies are significantly lower than the benchmark. The accuracy does gradually go up with increasing size of latent space, eventually approaching the benchmark when the latent space is close to the input size. This holds for all shallow learners.

5 Model Comparison & Interpretability

5.1 SHAP values

SHAP (SHapley Additive exPlanations) is a method for explaining machine-learning models based on Shapley values from cooperative game theory. The absolute SHAP value of a feature may be interpreted as its weight in model outputs.

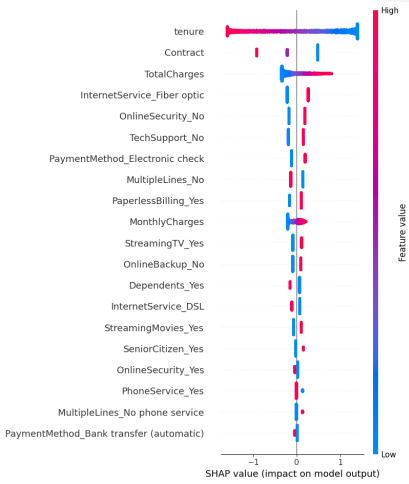


Figure 1: Logistic Regression SHAP summary plot

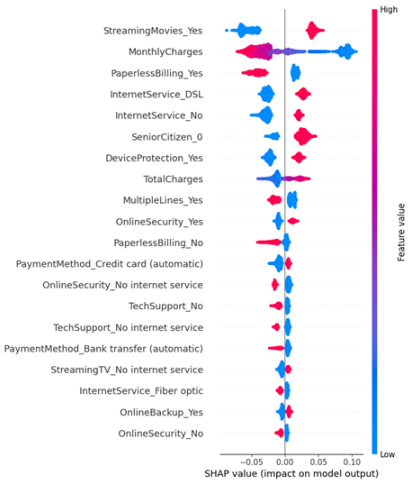


Figure 2: Random Forest SHAP summary plot

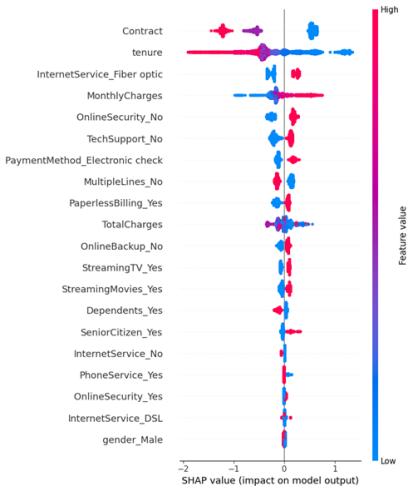


Figure 3: XGBoost SHAP summary plot

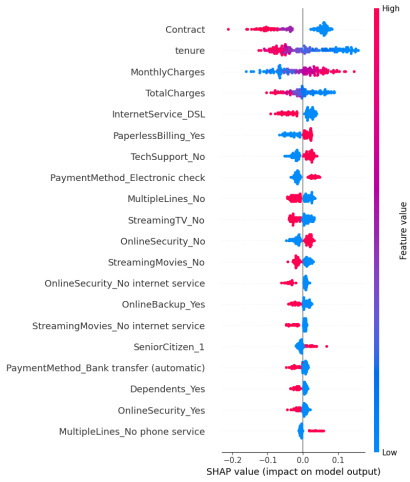


Figure 4: Feed-forward neural network SHAP summary plot

SHAP values make it possible to compare how different models assign importance to the same features, thereby highlighting model-specific differences in how predictions are formed. Across all models, MonthlyCharges and TotalCharges are identified as important features, although their relative rankings differ. Furthermore, Contract and Tenure are among the most influential features for Logistic Regression, XGBoost, and the neural network, but are less prominent for the Random Forest model. Notably, the Random Forest assigns high importance to StreamingMovies_Yes, a feature that none of the other models consider nearly as influential.

6 Conclusion

In this project we evaluated several models for churn prediction with a focus on the F_2 score, reflecting a business goal of prioritizing recall over precision for the minority class. Across all models, optimizing for F_2 increased recall at the cost of lower overall accuracy

Among all methods, XGBoost tuned for F_2 achieved the best balance between recall and precision and is therefore the most suitable model for deployment. Both logistic regression and random forest performed competitively indicating that limited model capacity is sufficient to achieve the upper limit of performance on this task.

Deep learning models did not meaningfully outperform shallow learners, possibly due to the limited dataset size. Autoencoder-based dimensionality reduction also failed to improve performance. SHAP analysis largely agreed with the EDA, and showed that *MonthlyCharges* and *TotalCharges* are important across all models, while *Contract* and *Tenure* are key drivers of churn for most models. Overall, classification performance

was driven by appropriate model choice and careful hyperparameter tuning rather than increased model complexity.